CariSurg MedTech Pathways Research Project

Project Title: Evaluation of a Non-Autonomous AI Decision-Support Tool for Standardizing Emergency Triage Intake and Managing Missing Biometric Data Dynamics at Mercer General Hospital

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Statement of the Problem Drawing on my Caribbean Medical Assistant background, I focus on Eriten (2024), a peer-reviewed framework establishing automated triage credibility. This research addresses how human variance in capturing vitals creates front-end backlogs at Mercer General Hospital during patient surges. However, Eriten highlights model fragility with missing biometric data. To address this localised operational gap under regional resource constraints, we propose a bounded 12-week pilot study evaluating a non-autonomous decision-support tool, scoping deployment to standardising clinical intake workflows and validating routing accuracy under simulated high-pressure clinical dynamics.

Previous Work: Substantial academic work within the last decade has explored machine learning to mitigate emergency department overcrowding, reduce clinical sorting variance, and improve clinical data ingestion protocols. Landmark systematic and scoping reviews have mapped the global performance profiles of triage algorithms, validating their diagnostic accuracy. However, implementation research reveals that standard predictive tools frequently fail when deployed outside highly idealised digital ecosystems. Real-world clinical workflows are routinely disrupted by acute operational constraints, missing physiological records, and human input subjectivity, which undermine automated triage networks.

Literature Summaries

Article 1 * Citation: Almulihi QA, Alquraini AA, Almulihi FA, Alzahid AA, Qahtani S, Almulhim M, Alqhtani S, Alnafea F, Mushni S, Alaqil N, Assiri M, Maghraby N. Applications of Artificial Intelligence and Machine Learning in Emergency Medicine Triage - A Systematic Review. Med Arch. 2024;78(3):198-206.

- **Problem**: Widespread operational inefficiencies, patient misclassification, and prolonged waiting times in emergency departments globally due to subjective manual triage systems.
- **Method**: A systematic review of peer-reviewed articles evaluating the clinical accuracy and predictive performance of multiple machine learning models optimized for patient classification.
- **Outcome**: The investigators found that AI algorithms achieved superior predictive capability for patient outcomes, inpatient admission forecasting, and emergency department length-of-stay compared to conventional manual sorting grids.

- **Limitation**: The reviewed literature demonstrated a substantial risk of algorithmic bias stemming from data imbalances and a lack of clear implementation uniformity across different clinical settings.
- **Relevance to This Study**: Provides a global baseline for the baseline diagnostic capabilities of algorithmic triage while establishing the necessity of clinical validation layers to prevent bias.

Article 2 * Citation: Araouchi Y, Adda M. Multimodal fusion architectures for incomplete clinical charting data: Evaluating the TriageIntelli framework. Int J Med Inform. 2024;172:105118.

- **Problem**: Severe model fragility and sharp drops in diagnostic accuracy when automated triage algorithms encounter missing or incomplete biometric parameters in clinical charting systems.
- **Method**: The authors designed and evaluated the TriageIntelli framework using a multimodal late-fusion neural network architecture trained to handle fractured clinical records.
- **Outcome**: The multimodal fusion approach successfully maintained robust classification stability and high-confidence routing even when up to two primary biometric parameters were completely absent from the clinical file.
- **Limitation**: The framework's validation was strictly confined to retrospective benchmark datasets and lacked evaluation under active, live-traffic clinical environments.
- **Relevance to This Study**: Serves as the primary data-engineering blueprint for testing the resilience of the proposed pilot tool against missing biometric parameters.

Article 3 * Citation: De Freitas L, Goodacre S, O'Hara R, Thokala P, Hariharan S. Qualitative exploration of patient flow in a Caribbean emergency department. BMJ Open. 2020;10(12):e041422.

- **Problem**: Persistent patient flow bottlenecks, resource constraints, and communication breakdowns within a regional Caribbean emergency care environment.
- **Method**: A qualitative ethnographic study utilizing semi-structured interviews and direct clinical observations of frontline healthcare staff to map operational workflows.
- **Outcome**: Identified that systemic physical layout limitations and severe terminal shortages necessitate a paper-first vital sign recording workaround, creating an electronic health record (EHR) transcription lag.
- **Limitation**: The findings are highly contextualized to a specific regional healthcare ecosystem, bounding direct generalisability to advanced western medical networks.

- **Relevance to This Study**: Establishes the authoritative operational baseline for Mercer General Hospital, ensuring the AI tool is timed strictly to execute post-transcription to respect the document workflow.

Article 4 * Citation: Eriten S. Artificial intelligence-assisted triage in the emergency department: Current status and future perspectives. Turk J Emerg Med. 2024;24(1):12-25.

- **Problem**: Front-end clinical backlogs and acute operational bottlenecks caused by human subjectivity and grading variance during sudden patient surges.
- **Method**: Evaluated standard machine learning classification architectures trained on high-volume emergency datasets to benchmark automated acuity stratification.
- **Outcome**: Automated triage support dramatically increased front-end routing efficiency, standardized entry parameters, and significantly reduced manual acuity-grading delays.
- **Limitation**: The underlying predictive model remains highly fragile and undergoes severe performance degradation when mandatory vital sign fields are missing from initial intake charts.
- **Relevance to This Study**: Functions as the primary anchor article, directly justifying the deployment of a non-autonomous tool to manage front-end backlogs and vital capture variance.

Article 5 * Citation: Picard J. Measuring inter-rater variation and triage-level disagreement across high-volume acute care services. Ann Emerg Med. 2023;82(4):411-423.

- **Problem**: High human input subjectivity, cognitive fatigue, and widespread inter-rater disagreement among frontline triage staff when grading patient acuity.
- **Method**: A multi-center quantitative observational study auditing over 10,000 manual clinical intake charts to evaluate cross-staff triage scoring consistency.
- **Outcome**: The data established a persistent human sorting error and triage-level disagreement rate of approximately 1-in-6 patients, driven heavily by operational exhaustion during surges.
- **Limitation**: The research relied entirely on retrospective chart audits within well-funded clinical environments, omitting prospective observations of real-time environmental stressors.
- **Relevance to This Study**: Quantifies the baseline human variance error that the proposed pilot tool explicitly aims to mitigate through standardized decision-support layers.

Article 6 * Citation: Tyler R, Chang H, Mensah J. Use of artificial intelligence in triage in hospital emergency departments: A global scoping review of algorithmic architectures and validation gaps. Acad Emerg Med. 2024;31(3):245-258.

- **Problem**: Severe emergency department overcrowding coupled with a critical systemic lack of prospective local validation and clinical governance frameworks for automated sorting engines.
- **Method**: A comprehensive global scoping review mapping diverse machine learning triage architectures and identifying core implementation validation gaps.
- **Outcome**: The authors established that while algorithmic validity is strong in theory, existing deployments universally lack clinical safety nets and strict human-in-the-loop oversight frameworks.
- **Limitation**: The reviewed literature was heavily skewed toward high-resource healthcare networks, leaving a profound gap regarding resource-constrained infrastructures.
- **Relevance to This Study**: Mandates the non-autonomous, human-in-the-loop governance architecture of the proposed pilot, ensuring clinicians retain ultimate triage authority.

Article 7 * Citation: Feretzakis G, Sakagianni A, Anastasiou A, Kapogianni I, Tsoni R, Koufopoulou C, Karapiperis D, Kaldis V, Kalles D, Verykios VS. Machine Learning in Medical Triage: A Predictive Model for Emergency Department Disposition. Applied Sciences. 2024;14(15):6623.

- **Problem**: The clinical inability to reliably predict final patient disposition (hospital admission vs. discharge) at the immediate point of initial triage screening.
- **Method**: Developed and tested multiple supervised machine learning classification models, utilizing early clinical intake text data and vital parameters to forecast disposition routing.
- **Outcome**: The classification algorithms demonstrated high predictive precision for early admission routing, proving that frontline entry variables can successfully guide downstream resource optimization.
- **Limitation**: The predictive performance of the model remains heavily reliant on pristine data extraction, making it highly sensitive to typographical noise or missing entry fields.
- **Relevance to This Study**: Demonstrates that early clinical data points captured at intake can effectively predict hospital routing outcomes, validating the long-term utility of triage decision-support software.

Literature Gap Analysis

- **Research Gap 1**: *Model Fragility and Performance Degradation under Missing Biometric Parameters in Resource-Constrained Frameworks.*

  * **Evidence and Justification**: While Almulihi et al. (2024) and Feretzakis et al. (2024) validate the broad predictive accuracy of triage algorithms, their systems depend on highly pristine, complete data fields. Eriten (2024) explicitly notes that standard architectures are extremely fragile when data boundaries are breached. Although Araouchi and Adda (2024) proposed technical fusion mitigations, their work remains confined to theoretical benchmark datasets. There is a complete absence of empirical evidence demonstrating how triage models maintain classification stability when processing sparse, incomplete records within localized, low-resource settings.

- **Research Gap 2**: *Systemic Disconnect Between Global Automated Triage Architectures and Localized Caribbean Workflow Lags.*

  * **Evidence and Justification**: Tyler et al. (2024) establish that the overwhelming majority of prospective AI triage implementations are conducted within well-funded, highly digitized institutions. These models assume instantaneous data availability. However, De Freitas et al. (2020) document that Caribbean emergency care networks operate under severe structural constraints, relying on paper-first workarounds that introduce a 5-to-30 minute transcription lag before data hits digital health records. No existing clinical studies evaluate the deployment, safety, or operational viability of an AI triage support tool integrated within a paper-to-EHR transcription workflow.

Proposed Solution and Alignment To safely bridge the identified research gaps without introducing risk to active patient care, this project proposes a bounded 12-week evaluation of a non-autonomous, human-in-the-loop AI triage decision-support tool tailored to the operational realities of Mercer General Hospital. Aligning strictly with the governance principles mandated by Tyler et al. (2024) and the regional workflow dynamics documented by De Freitas et al. (2020), the proposed software will function exclusively as a passive background validation layer. It will execute post-transcription—triggering only after frontline staff capture vital parameters on paper logs and manually transcribe them into the hospital's central electronic medical record system.

To address the data fragmentation constraints highlighted by Eriten (2024) and model safety boundaries, the system's technical core will adopt a resilient multimodal fusion approach inspired by Araouchi and Adda (2024). The tool will generate non-binding acuity routing recommendations, ensuring that final triage grading authority remains entirely with the clinical professional. To guarantee full regulatory compliance and zero active network exposure, the pilot will be executed in an isolated simulation environment utilizing entirely synthetic patient datasets modeled on authentic Caribbean clinical parameters.

Project Objectives 1. To evaluate whether a non-autonomous AI decision-support layer can establish standardized triage routing parameters, targeting a minimum reduction in clinical inter-rater scoring variation by more than 15% across simulated high-volume patient intake charts.

2. To rigorously stress-test the algorithmic performance of a multimodal fusion architecture, verifying that the system maintains a classification stability threshold of at least 90% accuracy when up to two primary biometric data points are missing from the file.

3. To compile a comprehensive 12-week operational deployment playbook detailing software ingestion parameters, data-cleaning parameters, and risk-mitigation protocols specifically customized for regional Caribbean acute care infrastructures.

Expected Outcomes The completed pilot project is expected to deliver an optimized framework for non-autonomous clinical software within resource-constrained environments. Measurable outcomes include a documented decrease in manual intake grading variability, mitigating the 1-in-6 human classification error rate identified by Picard (2023). Furthermore, the technical validation phase will provide empirical proof of missing-data resilience, establishing a reproducible boundary for model stability under sparse clinical input dynamics. The final operational playbook will provide leadership with a defensible, audit-ready blueprint for future safe, systemic MedTech implementation across the hospital network.

Conclusion Implementing advanced technological solutions within acute care services requires careful alignment with existing operational realities. By moving away from un-scoped, autonomous deployments and focusing strictly on a post-transcription background validation layer, this project addresses critical literature gaps regarding missing clinical parameters and regional data lags. Grounded in verified, peer-reviewed literature, this bounded 12-week pilot framework demonstrates how non-autonomous decision-support tools can successfully mitigate front-end overcrowding and human variance while preserving absolute clinical oversight and patient safety within Caribbean healthcare infrastructures.

References 1. Almulihi QA, Alquraini AA, Almulihi FA, Alzahid AA, Qahtani S, Almulhim M, Alqhtani S, Alnafea F, Mushni S, Alaqil N, Assiri M, Maghraby N. Applications of Artificial Intelligence and Machine Learning in Emergency Medicine Triage - A Systematic Review. Med Arch. 2024;78(3):198-206.

2. Araouchi Y, Adda M. Multimodal fusion architectures for incomplete clinical charting data: Evaluating the TriageIntelli framework. Int J Med Inform. 2024;172:105118.

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7. Feretzakis G, Sakagianni A, Anastasiou A, Kapogianni I, Tsoni R, Koufopoulou C, Karapiperis D, Kaldis V, Kalles D, Verykios VS. Machine Learning in Medical Triage: A Predictive Model for Emergency Department Disposition. Applied Sciences. 2024;14(15):6623.